

Social Problem Statement & Deliverables

Out-of-Pocket Healthcare Costs and Poverty: A Statistical Study of Patient Expenses in Kolhapur City

SOCIAL PROBLEM STATEMENT

Rising out-of-pocket (OOP) healthcare expenditures have become a critical socio-economic challenge for families living in Kolhapur city. As the costs of doctor consultations, prescription medicines, diagnostic tests, and hospital admissions escalate annually, households are placed under severe financial pressure. Because the vast majority of local residents pay for medical care entirely out of their own pockets without adequate health insurance coverage or public financial support, a single acute or chronic illness frequently threatens their economic stability.

This financial burden forces vulnerable groups, particularly lower-income households and families dependent on daily wages, to delay essential medical treatments, borrow capital at high-interest rates, or aggressively cut spending on fundamental human needs like food and education. Consequently, substantial healthcare spending acts as a primary catalyst for driving families into debt traps and pushing vulnerable populations back down below the poverty line. There is an urgent need to systematically measure these primary medical and non-medical costs, map their distribution across demographic strata, and understand the concrete impoverishing effects they exert on the local community.

"How can the department leverage primary patient spending data to identify healthcare expenditure trends, evaluate structural drivers of catastrophic medical debt, and optimize public financial protection strategies to protect vulnerable households from health-induced poverty in Kolhapur city?"

PROJECT DELIVERABLES

- 1. Data Preparation & Cleaning (Python):** Clean, code, and preprocess the raw primary survey data collected from 305 individuals in Kolhapur city. Use pandas and numpy to handle missing values, resolve anomalies, format variable types, and structure medical vs. non-medical expenses for statistical analysis.
- 2. Statistical Analysis & Modeling (SQL/Python/R):** Structure the cleaned patient data into a tabular relational format. Formulate descriptive indicators (mean, median, and percentages) and deploy advanced inferential statistical techniques—such as correlation, regression, and quantile evaluation—to isolate the primary drivers of catastrophic health expenditures and measure exact poverty status shifts.
- 3. Data Visualization & Dashboarding (Power BI / Python):** Construct an interactive analytics dashboard featuring demographic filters to visualize spending patterns. Highlight critical variations

between inpatient and outpatient costs, insured vs. uninsured individuals, and urban vs. rural cohorts to present risk concentrations visually.

4. **Comprehensive Project Report & Presentation:** Author a structured, academic-grade research report summarizing key statistical findings, methodology limitations, and data-driven policy recommendations. Prepare an accompanying presentation deck to communicate actionable insights effectively to healthcare stakeholders and policymakers.
 5. **GitHub Repository:** Compile and document all technical components in a clean, version-controlled repository, including data preprocessing scripts, SQL schema queries, dashboard files, and the finalized report documentation.
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